

Density dependence

Learning objectives

1. Understand how resource and space limitation and natural enemies affect population growth
2. Model density dependence with continuous time using a logistic model
3. Model density dependence with discrete time using a Beverton-Holt model and Ricker model

review of last lecture: structured populations, but population growth rate wasn't affected by abundance in any stage

now we're adding density dependence, population growth depends on abundance

Density-dependent processes

why would the number of individuals affect the growth of the population?

Resource limitation

- shared pool of resources *light, water, nutrients*
- as resource availability decreases, population growth decreases
- example: self-thinning
 - population size increases
 - intercept light from seedlings
 - seedling mortality increases

Space limitation

- can be independent of resource availability
- example: lateral growth *in tightly packed forests*, windthrow from belowground constraints

Natural enemies

- herbivores, pathogens, seed predators
- as population size increases, attracts herbivores and predators, susceptible to disease
- survival decreases, back to equilibrium

start with continuous time since more familiar: logistic equation

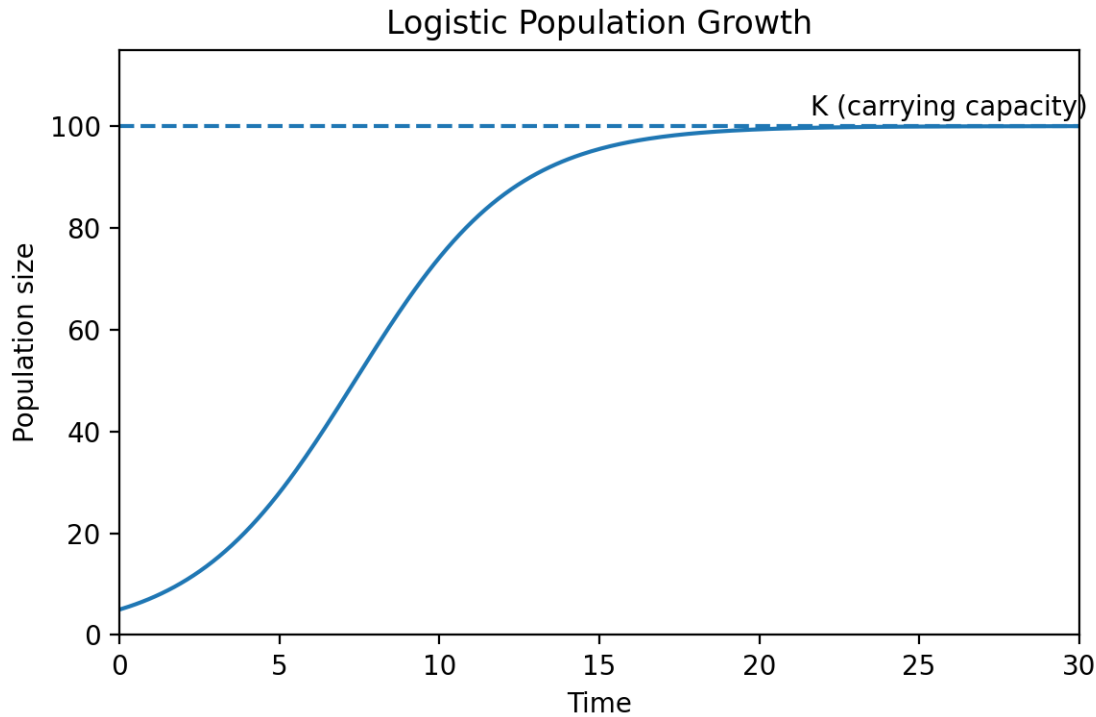
Logistic equation - continuous time

- r = intrinsic growth rate *same as r from first lecture, continuous time*
- K = carrying capacity *equilibrium abundance*

$$\frac{dN}{dt} = rN\left(1 - \frac{N}{K}\right)$$

change in population size over time is rN , what we saw before

now limited by carrying capacity, so if $N = K$, no change in size



K vs. α

K comes from early population ecology in fisheries, less useful to us

in plant populations, more interested in that α for determining strength of density dependence and competition

$$\alpha = \frac{r}{K}$$

$$\frac{dN}{dt} = N(r - \alpha N)$$

population size at time t = population size times the intrinsic growth rate minus density dependence

now let's go into some models for discrete time, which we'll play with in the lab and can be more common for analysis

two different models: Beverton-Holt and Ricker

Beverton-Holt model - discrete time

- λ - population growth rate
- α - phenomenological competition *strength of competition between conspecifics*

$$N_{t+1} = \frac{\lambda N_t}{1 + \alpha N_t}$$

again λN is that unrestricted population growth, but now it's limited by abundance with the competition scalar

Ricker model - discrete time

- same variables

$$N_{t+1} = \lambda N_t e^{-\alpha N_t}$$